

Practical-4

AIM: Study and Configuration of Virtual LAN (VLAN) using Cisco Packet Tracer.

➤ REQUIREMENTS

1. Any Computer or Laptop
2. Computer system with minimum 4 GB RAM
3. Cisco Packet Tracer software
4. Windows Operating System

➤ THEORY

A **Virtual Local Area Network (VLAN)** is a logical grouping of devices within a network regardless of their physical location. VLANs are used to divide a single physical network into multiple logical networks to improve security, performance, and management.

VLAN operates at the **Data Link Layer (Layer 2)** of the OSI model and is configured on switches.

Advantages of VLAN:

- Reduces broadcast traffic
- Improves network security
- Better network management
- Logical segmentation of departments (e.g., HR, Sales, IT)

In Cisco switches, VLANs are created using CLI commands.

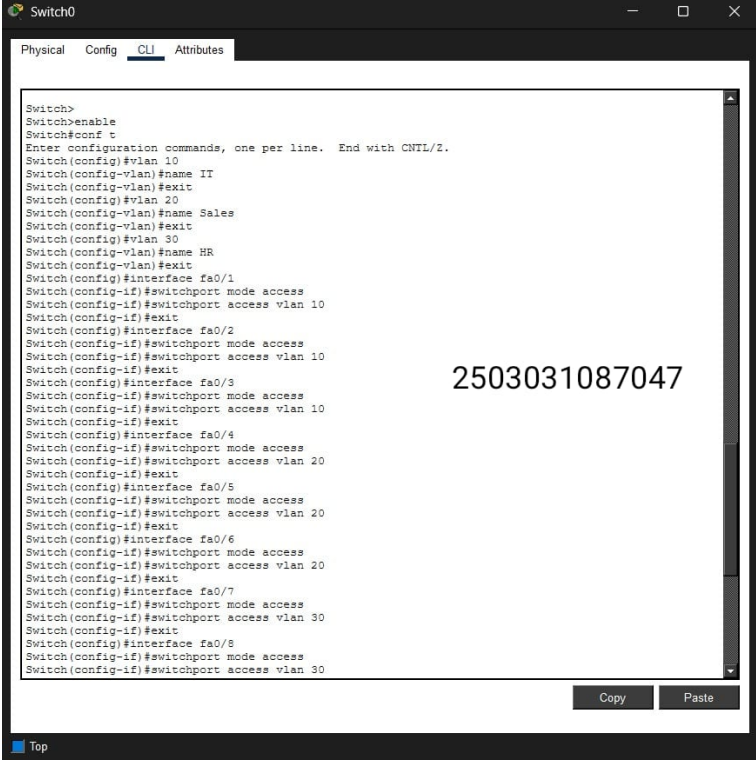
➤ PROCEDURE

- Open **Cisco Packet Tracer**.
- Create a new project.
- Place one **Switch (2960)** in the workspace.
- Add multiple **PCs** and connect them to the switch using straight-through cables.
- Assign IP addresses to PCs (Example):
 - VLAN 10 → 192.168.10.1 /24
 - VLAN 20 → 192.168.20.1 /24
 - VLAN 30 → 192.168.30.1 /24



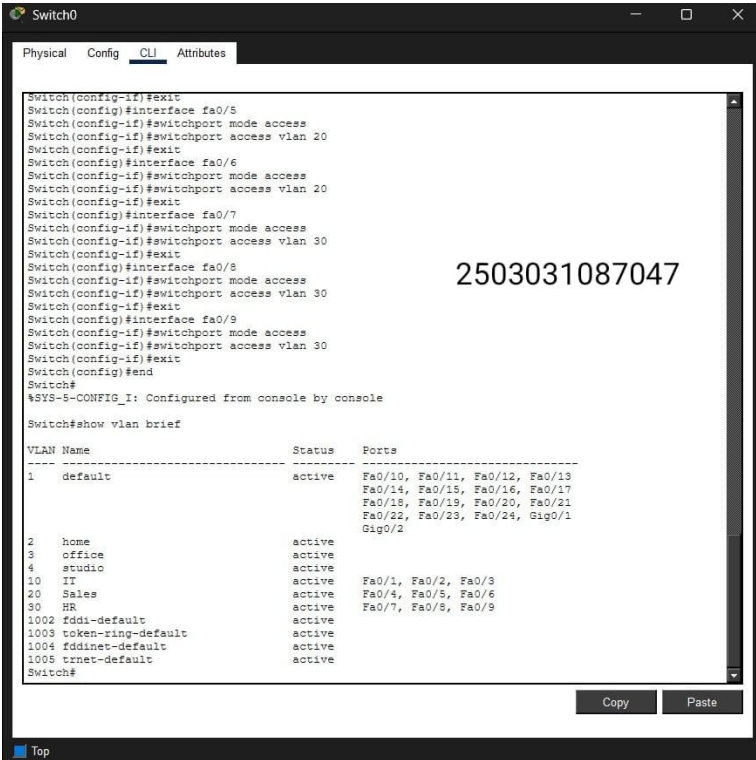
- **VLAN Configuration using CLI**
- Click on the switch → Go to **CLI mode**.
- Enter the following commands:
 - enable
 - configure terminal
 - vlan 10
 - name IT
 - exit
 - vlan 20
 - name Sales
 - exit
 - vlan 30
 - name HR
 - exit
- Assign ports to VLAN 10:
 - interface fa0/1
 - switchport mode access
 - switchport access vlan 10
 - exit
- Assign ports to VLAN 20:
 - interface fa0/4
 - switchport mode access
 - switchport access vlan 20
 - exit
- Assign ports to VLAN 30:
 - interface fa0/7
 - switchport mode access
 - switchport access vlan 30
 - exit
- Save configuration
- Test communication using **ping command**.
- Observe that PCs in same VLAN can communicate, but different VLANs cannot (without router).

➤ OUTPUT



```
Switch0
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
```

Figure 1: VLAN creation and port assignment using switch CLI configuration commands.



```
Switch0
Switch(config-if)#exit
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config)#interface fa0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
2    home                    active
3    office                  active
4    studio                  active
10   IT                       active    Fa0/1, Fa0/2, Fa0/3
20   Sales                   active    Fa0/4, Fa0/5, Fa0/6
30   HR                      active    Fa0/7, Fa0/8, Fa0/9
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Figure 2: Verification of configured VLANs using 'show vlan brief' command in Cisco Packet Tracer.

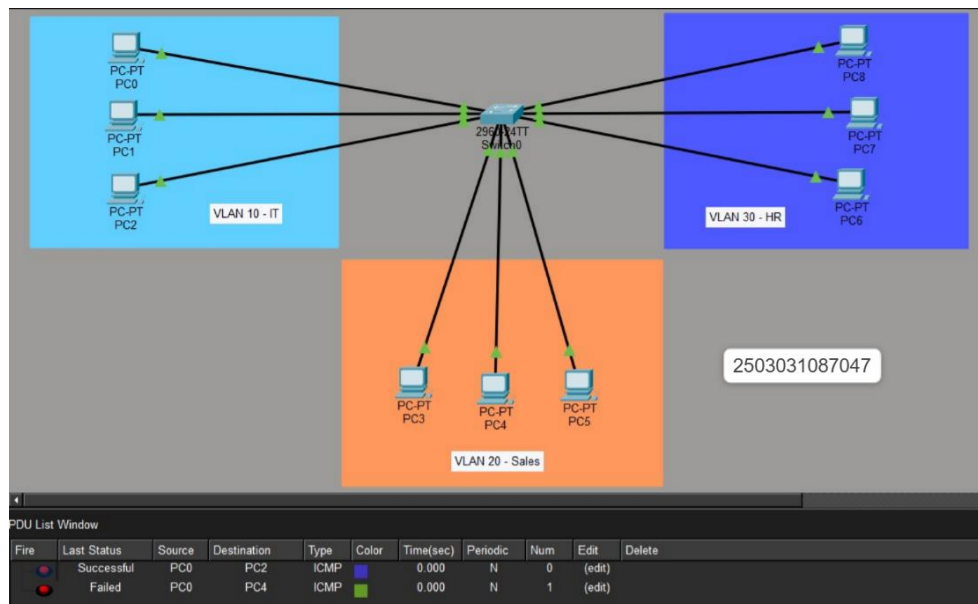


Figure 3: VLAN topology showing logical segmentation of IT (VLAN 10), Sales (VLAN 20), and HR (VLAN 30) departments in Cisco Packet Tracer.

➤ OBSERVATION

- VLAN 10 devices can communicate with each other.
- VLAN 20 devices can communicate with each other.
- Devices in different VLANs cannot communicate without inter-VLAN routing.
- VLAN reduces broadcast traffic and increases security.

➤ CONCLUSION

Thus, the study and configuration of Virtual LAN (VLAN) was successfully performed using Cisco Packet Tracer. It was observed that VLAN logically segments the network into separate broadcast domains, improving security, performance, and network management.